

NeuralChat

A Local LLM Chatbot — Project Report

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GitHub: <https://github.com/shakib1404/chatbot.git>

1. PROJECT SUMMARY

NeuralChat is a local LLM chatbot built with Streamlit for the UI, MongoDB for persistent storage, and Ollama for model inference. The app supports user registration and login, maintains per-user chat history, and sends recent conversation context to the model so replies remain coherent across turns.

2. ARCHITECTURE

- `app.py` — initializes Streamlit, loads session defaults, and routes to login or chat.
- `ui/auth.py` — handles registration, login, and session state changes.
- `ui/chat.py` — renders the chat interface, loads history, submits prompts, and streams responses.
- `db.py` — provides MongoDB helpers for users, messages, and statistics.
- `ollama_client.py` — wraps the Ollama REST API for normal and streaming responses.
- `styles.py` — injects custom dark-theme UI styling.

3. KEY FEATURES

Authentication	SHA-256 hashed passwords stored in MongoDB with unique username enforcement.
Chat Flow	History loaded from DB, rendered in order, new prompt stored as pending, then sent to Ollama with context.
Stop Generation	Stop button (on_click callback) sets a session flag; animation loop breaks on next iteration.
Streaming	Word-by-word animation with a blinking cursor; partial replies tagged and excluded from future context.
Data Storage	Two MongoDB collections: Users (username, password_hash, created_at) and Messages (username, role, content, timestamp UTC).
Configuration	OLLAMA_URL, OLLAMA_MODEL, OLLAMA_TIMEOUT_SECONDS, OLLAMA_CONTEXT_TURNS all centralized in config.py.

4. SETUP & RUN

1. `pip install -r requirements.txt`
2. Start MongoDB
3. `ollama serve && ollama pull mistral`
4. `streamlit run app.py`